

EcoView Production Deployment Guide

TL;DR — No Render/Railway needed. The app can run fully on Firebase (Auth + Firestore + Storage) + Vercel + Hugging Face Spaces. Zero paid services.

Architecture (Zero-Cost, No Server Required)

[React Frontend] Vercel (free)

- > **Firebase Auth** login, register, session tokens
- > **Firestore** all incident data, user profiles, leaderboard
- > **Firebase Storage** photo uploads
- > **Hugging Face Space** ML classification + spatial analytics

- > **Gemini API** VLM teacher fallback

What You Need From Scratch

Step 1 — Firebase Project Setup (10 min)

1. Go to console.firebase.google.com **Add Project**
2. Enable **Authentication** Sign-in methods enable **Email/Password**
3. Enable **Firestore Database** Start in production mode choose region nearest to your users (e.g. `asia-south1` for India)
4. Enable **Storage** Start in production mode
5. Go to **Project Settings** **Your Apps** **Web App** register app copy the config block

Your config will look like:

```
const firebaseConfig = {
  apiKey: "...",
  authDomain: "ecoview-xyz.firebaseio.com",
```

```
projectId: "ecoview-xyz",
storageBucket: "ecoview-xyz.appspot.com",
messagingSenderId: "123456789",
appId: "1:123456789:web:abc123"
};
```

6. Set **Firestore Security Rules** — go to Firestore Rules paste:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{uid} {
      allow read: if request.auth != null;
      allow write: if request.auth.uid == uid;
    }
    match /incidents/{id} {
      allow read: if true;
      allow create: if true;
      allow update: if request.auth != null && (
        resource.data.reporter_uid == request.auth.uid ||
        get(/databases/{database}/documents/users/{request.auth.uid}).data.role == 'admin'
      );
    }
  }
}
```

7. Set **Storage Rules** — go to Storage Rules paste:

```
rules_version = '2';
service firebase.storage {
  match /b/{bucket}/o {
    match /incidents/{allPaths=**} {
      allow read: if true;
      allow write: if request.resource.size < 10 * 1024 * 1024
        && request.resource.contentType.matches('image/.*');
    }
  }
}
```

Step 2 — Google Gemini API Key (2 min)

1. Go to aistudio.google.com
 2. Sign in — click **Get API Key** **Create API Key**
 3. Copy the key — you will use it in the next step as `GEMINI_API_KEY`
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Step 3 — Deploy ML Microservice to Hugging Face (15 min)

1. Sign up at huggingface.co
 2. Click + (top right) **New Space**
 3. Settings:
 - **Space name:** `ecoview-ml`
 - **SDK:** `Docker`
 - **Hardware:** `CPU Basic (Free)`
 - **Visibility:** `Public`
 4. Click **Create Space** — it creates a git repo
 5. Upload these files from `EcoView/ecoview-ml-backend/` into the Space repo (use the web UI or `git push`):
 - `Dockerfile`
 - `main.py`
 - `requirements.txt`
 - `core/` folder (all 4 files)
 - `spatial_analysis/` folder (all 3 files)
 6. Go to **Settings** tab in the Space **Variables and Secrets**:
 - Add **Secret:** `GEMINI_API_KEY` = [your key from Step 2]
 - Add **Variable:** `ALLOWED_ORIGINS` = `https://ecoview.vercel.app` (update after Step 4)
 7. Wait ~3 min for the Docker build to complete. Your Space URL will be:
`https://YOUR-USERNAME-ecoview-ml.hf.space`
 8. Test it: `https://YOUR-USERNAME-ecoview-ml.hf.space/health` should return
`{"status":"healthy",...}`
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Step 4 — Deploy Frontend to Vercel (5 min)

1. Push your code to GitHub (the `EcoView/` folder as the repo root)
2. Go to vercel.com **Add New Project** import your GitHub repo
3. Configure:
 - **Framework Preset:** `Vite`
 - **Root Directory:** `frontend`

- **Build Command:** `pnpm build`
- **Output Directory:** `dist`

4. Add **Environment Variables** (from your Firebase config in Step 1):

Variable	Value
VITE_FIREBASE_API_KEY	From Firebase project config
VITE_FIREBASE_AUTH_DOMAIN	e.g. <code>ecoview-xyz.firebaseio.com</code>
VITE_FIREBASE_PROJECT_ID	e.g. <code>ecoview-xyz</code>
VITE_FIREBASE_STORAGE_BUCKET	e.g. <code>ecoview-xyz.appspot.com</code>
VITE_FIREBASE_MESSAGING_SENDER_ID	From Firebase config
VITE_FIREBASE_APP_ID	From Firebase config
VITE_ML_URL	Your Hugging Face Space URL from Step 3

5. Click **Deploy** your frontend is live at `https://ecoview.vercel.app` (or similar)
6. Go back to Hugging Face Space Settings update `ALLOWED_ORIGINS` with your actual Vercel URL

What Is NOT Needed (can ignore)

Service	Why not needed
Supabase / PostgreSQL	Firebase Firestore replaces it
Redis (Upstash)	No caching layer needed at this scale
Cloudflare R2	Firebase Storage replaces it
Render / Railway / Fly.io	No backend server needed
Firebase Admin SDK	Only needed server-side — unused now

Final Checklist Before Going Live

- ☐ Firebase Auth enabled (Email/Password)
- ☐ Firestore database created with security rules above
- ☐ Firebase Storage enabled with image rules above

- ☐ Gemini API key obtained
- ☐ Hugging Face Space deployed and `/health` returns 200
- ☐ Vercel project deployed with all 7 env vars set
- ☐ `ALLOWED_ORIGINS` on HF Space updated to Vercel URL
- ☐ Test: register login submit report verify map shows pin